BUILDING AN INFINITE COMPATIBLE SEQUENCE OF THICKENINGS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3175-3208 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 3210-3254, the complete author proof (43 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 25 of queue_artin.json (pb=1644, stmt 3175-3208, proof 3210-3254).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned. Named omission: 'Some details omitted' for the transfer of property (3) - routine shrinking bookkeeping.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0G2J: sources/stacks-artin/pinned-original-source.tex, source lines 2951–2972. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07RT: sources/stacks-artin/pinned-original-source.tex, source lines 3254–3254. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let $\mathcal{X}$ be a category fibred in groupoids over $(Sch/S)_{fppf}$ having (RS*). Let x be an object of $\mathcal{X}$ over an affine scheme U of finite type over S. Let $u_n \in U$, $n \geq 1$ be a finite type points such that (a) there are no specializations $u_n \rightsquigarrow u_m$ for $n \neq m$, and (b) x is not versal at u_n for all n. Then there exist morphisms*

$$x \to x_1 \to x_2 \to \ldots \quad \text{in } \mathcal{X} \text{ lying over} \quad U \to U_1 \to U_2 \to \ldots$$

over S such that

(1) *for each n the morphism $U \to U_n$ is a first order thickening,*

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *for each n we have a short exact sequence*
$$0 \to \kappa(u_n) \to \mathcal{O}_{U_n} \to \mathcal{O}_{U_{n-1}} \to 0$$
with $U_0 = U$ for $n = 1$,

(3) *for each n there does* **not** *exist a pair (W, α) consisting of an open neighbourhood $W \subset U_n$ of u_n and a morphism $\alpha : x_n|_W \to x$ such that the composition*
$$x|_{U \cap W} \xrightarrow{\text{restriction of } x \to x_n} x_n|_W \xrightarrow{\alpha} x$$
is the canonical morphism $x|_{U\cap W} \to x$.

Proof. Since there are no specializations among the points u_n (and in particular the u_n are pairwise distinct), for every n we can find an open $U' \subset U$ such that $u_n \in U'$ and $u_i \notin U'$ for $i = 1, \ldots, n-1$. By Lemma 0G2J for each $n \geq 1$ we can find
$$x \to y_n \quad \text{in } \mathcal{X} \text{ lying over } \quad U \to T_n$$
such that

(1) the morphism $U \to T_n$ is a first order thickening,
(2) we have a short exact sequence
$$0 \to \kappa(u_n) \to \mathcal{O}_{T_n} \to \mathcal{O}_U \to 0$$
(3) there does **not** exist a pair (W, α) consisting of an open neighbourhood $W \subset T_n$ of u_n and a morphism $\beta : y_n|_W \to x$ such that the composition
$$x|_{U \cap W} \xrightarrow{\text{restriction of } x \to y_n} y_n|_W \xrightarrow{\beta} x$$
is the canonical morphism $x|_{U\cap W} \to x$.

Thus we can define inductively
$$U_1 = T_1, \quad U_{n+1} = U_n \amalg_U T_{n+1}$$
Setting $x_1 = y_1$ and using (RS*) we find inductively x_{n+1} over U_{n+1} restricting to x_n over U_n and y_{n+1} over T_{n+1}. Property (1) for $U \to U_n$ follows from the construction of the pushout in More on Morphisms, Lemma 07RT. Property (2) for U_n similarly follows from property (2) for T_n by the construction of the pushout. After shrinking to an open neighbourhood U' of u_n as discussed above, property (3) for (U_n, x_n) follows from property (3) for (T_n, y_n) simply because the corresponding open subschemes of T_n and U_n are isomorphic. Some details omitted. □